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Abstract

Aims

We aimed to determine whether treatment related changes in surrogate markers predict longer-term therapeutic effects on all-cause mortality and heart failure (HF) hospitalization in randomized trials of HF.

Methods

Systematic literature search included randomized trials of interventions in patients with HF until October 2024. Random-effects meta-regression models with inverse variance weighting were calculated between fifteen surrogate markers and the clinical endpoints of (i) all-cause mortality and (ii) HF hospitalization. The degree of heterogeneity of each model explained by the surrogate marker was determined with R^2 . Surrogate threshold effects (STEs) were also calculated.

Results

Ninety-six randomized trials with median follow-up 12 months were included, enrolling a total of 120 304 patients with HF. Treatment related changes in natriuretic peptides (NPs) were weakly correlated with all-cause mortality (64 comparisons, $P = .002$, $R^2 = 12\%$, and $STE = -34\%$) and HF hospitalization (41 comparisons, $P < .001$, $R^2 = 32\%$, and $STE = -30\%$). Changes in LVEF were moderately correlated with mortality (69 comparisons, $P < .001$, $R^2 = 59\%$, and $STE = +6\%$) and strongly correlated with HF hospitalization (45 comparisons, $P < .001$, $R^2 = 90\%$, and $STE = +2\%$), although this finding was only observed in trials of HFrEF. LV end-diastolic diameter was correlated with HF hospitalization but not mortality. Other echocardiographic markers were not predictive of clinical endpoints. Treatment related changes in patient reported outcomes and exercise capacity were either weakly correlated with clinical outcomes or derived from small clinical trials.

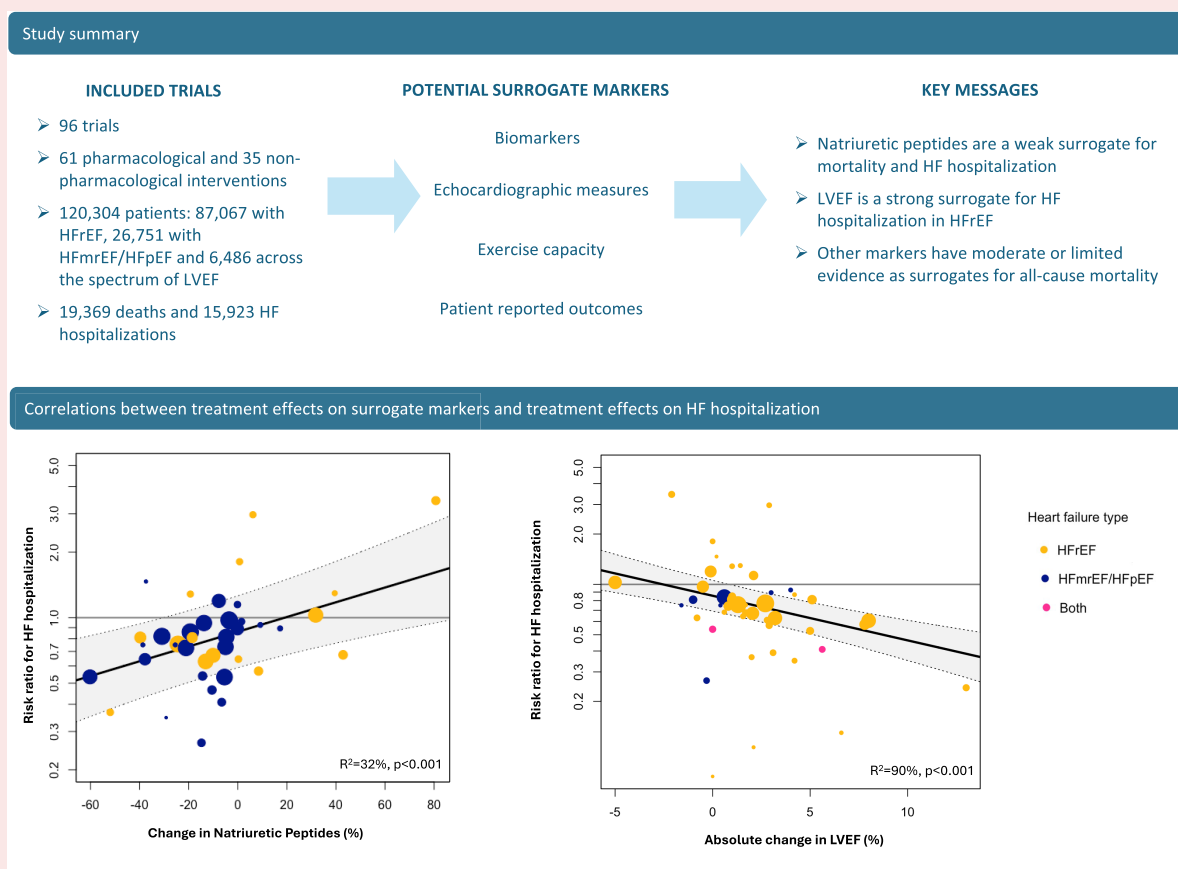
Conclusion

In HF trials, most surrogate markers, including NPs and echocardiographic measures were only weakly or moderately correlated with treatment effects on all-cause mortality. There is insufficient evidence to support the use of a single biomarker or echocardiographic measure in regulatory trials of HF.

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Graphical abstract



Surrogate markers of clinical endpoints in clinical trials of heart failure. Two example meta-regression plots are shown with each circle representing a trial, and the size of the circles are proportional to the inverse variance. Black straight lines represent weighted regression lines and dashed lines represent 95% prediction intervals. The grey straight line represents the line of unity where risk ratio = 1.0. HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; LVEF, left ventricular ejection fraction.

Keywords

Clinical trials • Heart failure • Surrogate outcomes

Introduction

Despite major advances in therapeutic options for HF over the last three decades, 5-year mortality rates remain ~50%.¹ The development of new therapeutics is hamstrung by rising costs in large part because regulatory approval still mostly relies on placebo (or sham) controlled design with demonstrated superiority in mortality and/or HF events.² Such clinical trials require very large sample sizes and long treatment durations for adequate statistical power. There is growing interest in identifying surrogate markers for these major clinical outcomes to gain regulatory approval and thereby improve the efficiency of drug development.

An ideal surrogate marker for clinical endpoints need to both (i) be causally linked to the clinical endpoint and (ii) capture the whole treatment effect upon the clinical endpoint.^{3,4} Although patient level studies can provide evidence that a surrogate and clinical endpoint are causally linked, trial-level validation is needed to assess whether a surrogate captures a large enough proportion of the treatment effect on the clinical endpoint.^{3,4} All-cause mortality represents the most objective and clinically meaningful outcome and captures the totality of treatment effects, including both direct therapeutic benefits and potential indirect

harms that may increase non-cardiovascular death. Demonstration of mortality reduction also provides the strongest evidence for drug approval.⁵ A recent review identified that most surrogate markers used as primary endpoints in clinical trials to support Food and Drug Administration (FDA) approval of non-oncologic chronic diseases lacked high-strength evidence of associations with clinical outcomes.⁶

Potential surrogate markers for HF include readily measurable variables such as echocardiographic measurements, natriuretic peptides (NPs), markers of exercise capacity, and patient reported outcomes. Whether changes in each of these variables can be used as surrogate endpoints for HF outcomes remains unclear. A previous study reported markers of left ventricular (LV) remodelling had modest correlations with mortality; however, this analysis relied on separate sets of trials to evaluate LV remodelling and mortality.⁷ Another meta-analysis found changes in NPs were not correlated with mortality and only modestly correlated with HF hospitalization.⁸ Furthermore, a separate study found exercise induced changes in exercise capacity measured with either peak oxygen uptake (VO_2) or 6-minute walk test (6MWT) were poor surrogate endpoints for mortality and HF hospitalization.⁹ However, these studies were likely underpowered because they included a limited number of trials with a smaller range of

treatment effects (need for both positive and negative trials). Therefore, the present systematic review and meta-analysis aims to assess the association between treatment related changes in surrogate markers and treatment effects on clinical endpoints of all-cause mortality and HF hospitalization in randomized clinical trials of HF.

Methods

Search strategy and selection criteria

We conducted a systematic review and meta-analysis according to the Preferred Reporting Items for Systematic Reviews and Meta analyses and Reporting of Surrogate Endpoint Evaluation using Meta analyses guidelines.^{10,11} The study was registered on PROSPERO CRD42023390653. We systematically searched Pubmed, Medline, and references of included trials and prior reviews for randomized clinical trials evaluating therapeutic interventions in patients with HF until October 2022, with the search updated until October 2024. The search terms included 'heart failure (HF)', 'NP', 'echocardiography', and 'randomized controlled trial'. Full search strategy can be found in the [Supplementary appendix](#). Inclusion criteria were randomized trials with (i) enrolled participants with established HF or cardiomyopathy, (ii) at least 100 patients, (iii) at least 6 months of follow-up duration, (iv) reported data on either all-cause mortality or HF hospitalization, and (v) reported data on change in at least one surrogate marker of interest. Trials of heart failure with reduced ejection fraction (HFrEF), mid-range ejection fraction (HFmrEF) and preserved ejection fraction (HFpEF) were included. Trials of non-HF patients, heart transplant populations and cardiac amyloidosis were excluded.

Screening and data collection

All identified studies were screened for inclusion in duplicate by two independent reviewers (T.H., S.I., N.R., R.A., and P.R.), first at the title and abstract level, and then at the full text level. Discrepancies were resolved after discussion with a senior author (N.W.). We extracted data on trial and participant characteristics, randomized treatments, baseline and reported mean difference for the change in surrogate marker and number of clinical endpoints according to treatment group. We recorded the between treatment group difference in surrogate marker when available, or alternatively we calculated the difference in change in mean surrogate marker from baseline to follow-up between treatment groups. We relied on trial-level estimates of treatment effects on surrogate markers and clinical endpoints. For assessments of change in NPs and left ventricular ejection fraction (LVEF), we extracted data at each available time point for time specific analyses because of the availability of such data. For other surrogates, we chose the time point closest to 12 months where possible. Populations were divided into HFrEF and HFmrEF/HFpEF according to a LVEF < or ≥40%, respectively. For trials with more than one intervention, we followed the Cochrane Handbook's approach for dealing with multi-arm trials.¹²

We examined surrogacy for four types of variables: (i) NPs (both N-terminal proBNP (NT-proBNP) and B-type NP (BNP)), (ii) echocardiographic measures, (iii) patient reported outcomes, and (iv) exercise capacity. The full list of variables include change in NPs; echocardiographic markers including change in LVEF, LV end-diastolic volume, LV end-systolic volume, LV end-diastolic diameter, LV end-systolic diameter, LV mass, left atrial volume, mitral annular average e' , and E/e' ; HRQoL scores assessed using Minnesota Living with Heart Failure Questionnaire (MLWHQ) or Kansas City Cardiomyopathy Questionnaire Overall Summary Score (KCCQ-OSS); exercise capacity measured by peak VO_2 and 6MWT. We chose KCCQ-OSS because it captures all components of the KCCQ across symptoms, functional limitations and quality of life.¹³ The primary clinical endpoint was all-cause mortality and the secondary clinical endpoint was HF hospitalization. The Cochrane Collaboration's risk of bias tool was used to determine study methodological quality.¹⁴

Statistical analysis

For each trial, we calculated the effect of the intervention vs control on clinical outcomes, expressed as log risk ratio (RR) and 95% confidence intervals, where $\text{RR} < 1.0$ indicated a beneficial effect of the intervention compared to control. If a study contributed to more than one treatment comparison, the

number of events and sample size in the shared group are then proportionally reduced for each split comparison. Surrogacy was assessed on the correlation between difference in change in surrogate marker between treatment groups and the magnitude of the treatment effect on clinical outcome using a random effects regression model with inverse variance weighting. We restricted analyses to complete cases and accounted for missing data using inverse variance weighting.¹⁵ Regression P -values < .05 were considered statistically significant and degree of heterogeneity of the model explained by the surrogate marker (R^2) were calculated with $R^2 \geq 70\%$, 50%–69%, and <50% representing strong, moderate and weak correlations, or low, moderate, and high degrees of heterogeneity, respectively.¹⁰ Regression models were constructed for each pair of surrogate marker and clinical endpoint that had at least five trials/comparisons. Subgroup analyses were also performed according to (i) whether the population included patients with HFrEF or HFmrEF/HFpEF and (ii) NT-proBNP or BNP for analyses of NPs. Finally, to identify the most informative monitoring interval for NPs and LVEF, we performed analyses according to the follow-up time point for reassessing change in NP/LVEF.

Among variables that were significantly associated with the clinical outcome (regression P -values < .05), we assessed surrogacy based on the following: (i) Surrogate threshold effect (STE), defined as the intercept of the prediction band of the regression line with zero effect on the clinical outcome and indicates the minimum change in surrogate marker to have 95% confidence in predicting a non-null effect on RRs for the clinical outcome.¹⁶ STE is distinct from the minimal clinically important difference, and reflects the mean change in a surrogate marker needed in a clinical trial to predict a corresponding reduction in mortality or HF hospitalization. (ii) Multivariate adjusted regression analyses by entering the following co-variables into the model—study publication year, treatment type (medical or non-medical intervention), HF type (either HFrEF or HFmrEF/HFpEF), placebo controlled (yes or no), median follow-up for the clinical outcome assessment, and time point for the measure of change in surrogate marker. (iii) Sensitivity analysis performed by weighting the regression models by sample size and by limiting analyses to trials with at least 1000 patient-years of follow-up. (iv) Leave-one-out cross validation analyses to validate results obtained in the main analysis. Each trial was left out one at a time, and the surrogate model was built with the other trials, this model was then reapplied to the left out trial to predict the effect of treatment on clinical endpoints. The leave-one-out cross validated R^2 was calculated as the correlation between the individual predictions made by the model over all trials and the actual treatment effects. We gave an overall grade for the surrogacy of each measure and clinical outcome as follows: limited/no evidence when there was weak or no correlation between the treatment related change in surrogate marker and change in clinical outcome or insufficient data to construct the regression model; moderate evidence when there was a statistically significant association between the treatment related change in surrogate marker and change in clinical outcome with moderate correlation ($R^2 = 50\%$ –69%) or strong correlation ($R^2 \geq 70\%$) but data were derived from small randomized trials with fewer than 1000 patient-years; strong evidence when there was a statistically significant association between the treatment related change in surrogate marker and change in clinical outcome with strong correlation ($R^2 \geq 70\%$) and analyses were robust after excluding small studies. All analyses were performed with R version 4.1.2.

Results

Included studies

Overall, 96 trials enrolling 120 304 patients with HF were included in this analysis ([Supplementary Figure S1](#) and [Table S1](#)); 69 trials included 87 067 patients with HFrEF, 19 trials included 26 751 patients with HFmrEF/HFpEF, and 8 trials included 6486 patients across the spectrum of ejection fraction. The 96 trials contributed to 108 comparisons, with 61 trials investigating pharmacological therapies, whilst 35 trials investigated non-pharmacological therapies including devices, interventional procedures, exercise regimens, supplementations, and dietary modifications ([Supplementary Table S1](#)). The median trial enrolment was 12 months (25th to 75th percentile: 8 to 24 months). There was a total of 19 369 deaths among 118 238 participants and 15 923

HF hospitalizations among 86 877 participants. 79 trials were assessed as low risk of bias, whilst the remainder as moderate risk (Supplementary Table S2). The number of participants and time points for assessment of each surrogate marker is shown in Supplementary Table S3.

Natriuretic peptides

Among 64 comparisons, there was significant association between treatment related percentage change in NPs and treatment

related effects on all-cause mortality ($P = .002$), although correlation was weak ($R^2 = 12\%$) (Figure 1A). Among trials of HFrEF, percentage change in NPs was weakly correlated with all-cause mortality (44 trials, $P = .001$, and $R^2 = 9\%$) and HF hospitalization (25 trials, $P = .001$, and $R^2 = 41\%$) (Supplementary Figure S2). Among trials of HFmrEF/HFpEF, percentage change in NPs were not associated with changes in mortality nor HF hospitalization. These findings were consistent when studies were limited to 43 comparisons using NT-proBNP ($P = .043$ and $R^2 = 37\%$) and 21 comparisons using BNP ($P = .017$ and $R^2 = 33\%$) (Supplementary Table S4). Findings were also consistent in multivariate regression analyses (64 comparisons, $P = .001$, and $R^2 = 47\%$), when weighting by sample size (64 comparisons, $P = .004$, and $R^2 = 12\%$) and when absolute change in NPs was used (Supplementary Tables S4 and S5). The correlation between NPs and all-cause mortality was weak when restricting analyses to trials with >1000 patient-years of follow-up (17 comparisons, $P = .173$, and $R^2 = 4\%$) (Supplementary Table S6). Among 41 comparisons, treatment related percentage change in NPs were associated with treatment effects on HF hospitalization, although correlation was also weak ($P < .001$ and $R^2 = 32\%$) (Figure 1B). The STE for change in NPs was -37% for all-cause mortality and -30% for HF hospitalization (Table 1). The optimal monitoring interval for NPs was six months, which correlated well with both all-cause mortality (33 comparisons, $P < .001$, and $R^2 = 100\%$) and HF hospitalization (19 comparisons, $P < .001$, and $R^2 = 74\%$) (Supplementary Table S4).

Echocardiographic markers

Therapeutic effects on LVEF were moderately correlated with treatment related effects on all-cause mortality (69 comparisons, $P < .001$, and $R^2 = 59\%$) (Figure 2 and Table 1). In multivariate regression analyses, changes in LVEF remained independently associated with effects on both mortality and HF hospitalization, but correlation was poor ($R^2 < 50\%$). Similar findings were observed when weighting by sample size and when limiting analyses to those trials with >1000 patient-years of follow-up (Supplementary Tables S4, S5 and S6). Subgroup analyses showed change in LVEF was associated with mortality in trials of HFrEF (56 comparisons, $P < .001$, and $R^2 = 49\%$) but not in trials of HFmrEF/HFpEF (11 comparisons, $P = .355$, and $R^2 = 0\%$) (Supplementary Table S4). In trials of HFrEF, treatment related changes in LVEF were strongly correlated with effects on HF hospitalization (36 comparisons, $P < .001$, and $R^2 = 86\%$) (Supplementary Figure S3). The STE for change in LVEF was an absolute increase of 8% for all-cause mortality and 2% for HF hospitalization (Table 1). The optimal monitoring interval for assessing change in LVEF was 12 months (Supplementary Table S4). Leave-one-out cross validation analyses for change in LVEF were broadly consistent (Supplementary Tables S7 and S8).

Except for LVEF and LV end-diastolic volume, no other echocardiographic measure was significantly associated with treatment effects on mortality (Table 1). Treatment related changes in LV end-diastolic volume was moderately correlated with risk of all-cause mortality (38 comparisons, $P = .009$, $R^2 = 41\%$, and $STE = -14\%$) (Figure 3). Treatment related changes in LV end-systolic volume and LV end-diastolic diameter were associated with treatment related changes in HF hospitalization (LV end-systolic volume: 19 comparisons, $P = .009$, $R^2 = 60\%$; LV end-diastolic diameter: 10 comparisons, $P = .004$, and $R^2 = 98\%$) (Supplementary Figure S4; Table 1). Among 6 trials of patients with HFmrEF/HFpEF, change in average mitral annular e' was significantly associated with risk of HF hospitalization (six comparisons, $P = .042$, $R^2 = 100\%$) but analyses were not robust after exclusion of small studies (Figure 3; Supplementary Table S6). Other echocardiographic measures were not significantly associated with HF hospitalization.

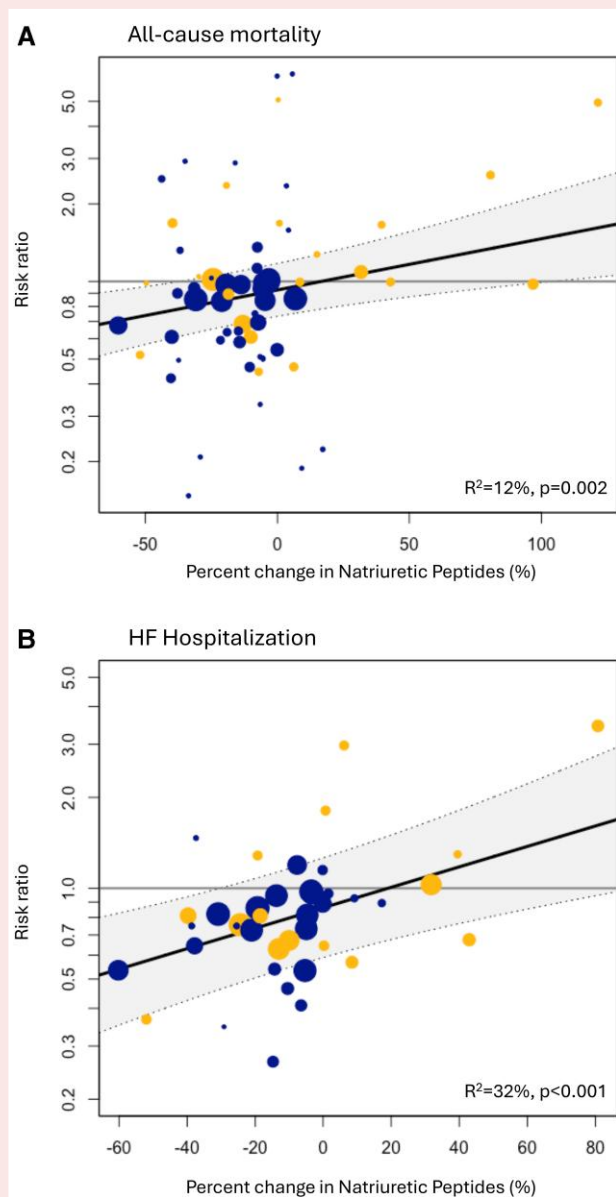


Figure 1 Correlations between treatment related effects on percentage change in natriuretic peptides and treatment effects on (A) all-cause survival and (B) HF hospitalization. Blue circles = NT-proBNP and yellow circles = BNP. Each circle represents a trial, and the size of the circles is proportional to the inverse variance. Black straight lines represent weighted regression lines and dashed lines represent 95% prediction intervals. The grey straight line represents the line of unity where $RR = 1.0$

Table 1 Association between surrogate markers and all-cause mortality and HF hospitalization

Surrogate marker	All-cause mortality					HF hospitalization				
	No. of comparisons analysed	Slope of regression line	P-value	R ²	Surrogate threshold effect	No. of comparisons analysed	Slope of regression line	P-value	R ²	Surrogate threshold effect
Biomarker										
Percentage change in Natriuretic peptides (%)										
HF+EF	44	0.005	.001	9%	−38%	25	0.008	.001	41%	−39%
HFmrEF/HFpEF	16	0.003	.56	0%	NA	12	0.007	.36	0%	NA
Overall	64	0.005	.002	12%	−37%	41	0.008	<.001	32%	−30%
Echocardiography										
Absolute change in LVEF (%)										
HF+EF	56	−0.057	<.001	49%	+8%	36	−0.068	<.001	86%	+2%
HFmrEF/HFpEF	11	−0.042	.36	0%	NA	7	0.086	.54	0%	NA
Overall	69	−0.049	<.001	59%	+6%	45	−0.061	<.001	90%	+2%
LV end-diastolic volume (%)										
HF+EF	31	0.030	.02	36%	−18%	19	0.019	.20	14%	NA
HFmrEF/HFpEF	6	0.112	.12	0%	NA	NA	NA	NA	NA	NA
Overall	38	0.033	.009	41%	−14%	23	0.020	.21	9%	NA
LV end-systolic volume (%)										
HF+EF	29	0.012	.08	41%	NA	18	0.022	.02	51%	−12%
HFmrEF/HFpEF	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Overall	32	0.012	.06	48%	NA	19	0.021	.009	60%	−9%
LV end-diastolic diameter (mm)										
HF+EF	12	0.104	.10	100%	NA	6	0.190	.001	100%	−0.6 mm
HFmrEF/HFpEF	6	−0.083	.82	0%	NA	NA	NA	NA	NA	NA
Overall	19	0.086	.12	100%	NA	10	0.150	.004	98%	−0.3 mm
LV end-systolic diameter (mm)										
HF+EF	7	0.130	.09	100%	NA	NA	NA	NA	NA	NA
HFmrEF/HFpEF	4	1.316	.18	0%	NA	NA	NA	NA	NA	NA
Overall	12	0.108	.08	100%	−0.8 mm	6	0.078	.56	0%	−1.3 mm
LV mass (%)										
HF+EF	8	0.018	.72	0%	NA	NA	NA	NA	NA	NA
HFmrEF/HFpEF	11	−0.025	.35	0%	NA	6	0.071	.43	6%	NA
Overall	19	0.004	.82	0%	NA	10	0.036	.21	48%	−0.8%
E/A ratio										
HF+EF	6	−0.180	.92	0%	NA	NA	NA	NA	NA	NA
HFmrEF/HFpEF	7	−4.531	.25	0%	NA	NA	NA	NA	NA	NA
Overall	13	−0.045	.97	0%	NA	6	1.982	.27	0%	NA

Continued

Table 1 Continued

Surrogate marker	All-cause mortality					HF hospitalization				
	No. of comparisons analysed	Slope of regression line	P-value	R ²	Surrogate threshold effect	No. of comparisons analysed	Slope of regression line	P-value	R ²	Surrogate threshold effect
LA volume (%)	11	−0.007	.68	0%	NA					
HFREF	5	−0.022	.12	100%	NA	NA	NA	NA	NA	NA
HFmrEF/HFpEF	6	0.082	.13	0%	NA	5	0.063	.09	47%	NA
Overall	11	−0.007	.68	0%	NA	8	0.018	.14	7%	−4%
Mitral annular e'										
HFREF	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
HFmrEF/HFpEF	8	1.004	.59	0%	NA	6	2.666	.04	100%	+0.2
Overall	11	1.191	.50	0%	NA	6	2.666	.04	100%	+0.2
E/e'										
HFREF	5	−1.012	.24	0%	NA	NA	NA	NA	NA	NA
HFmrEF/HFpEF	9	−0.049	.56	0%	NA	7	0.021	.70	0%	NA
Overall	14	−0.059	.49	0%	NA	9	0.022	.66	0%	NA
Patient reported outcomes										
MLWHFQ score										
HFREF	17	0.041	.052	0%	NA	9	0.036	.067	100%	NA
HFmrEF/HFpEF	6	0.009	.730	98%	NA	NA	NA	NA	NA	NA
Overall	24	0.041	<.001	93%	−2.6 points	13	0.056	<.001	100%	+0.2 points
KCCQ-OSS										
HFREF	9	−0.024	.0290	42%	+6 points	9	−0.018	.42	0%	NA
HFmrEF/HFpEF	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Overall	14	−0.025	.01	36%	+5 points	14	−0.023	.18	2%	NA
Exercise capacity										
6-minute walk test										
HFREF	21	−0.006	.01	77%	+34 m	13	−0.010	.01	75%	+38 m
HFmrEF/HFpEF	11	−0.022	.16	0%	NA	7	−0.050	.04	0%	+9 m
Overall	33	−0.006	.01	57%	+41 m	21	−0.010	.03	38%	+44 m
Peak VO ₂ ml/kg/min										
HFREF	7	−0.165	.06	46%	+0.9 ml/kg/min	NA	NA	NA	NA	NA
HFmrEF/HFpEF	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Overall	11	−0.167	.06	0%	+0.9 ml/kg/min	6	−0.298	.07	0%	NA

KCCQ-OSS, Kansas City Cardiomyopathy Questionnaire Overall Summary Score; LA, left atrial; LV, left ventricular; LVEF, left ventricular ejection fraction; m, metres; MLWHFQ, Minnesota Living with Heart Failure Questionnaire; VO₂ oxygen uptake.

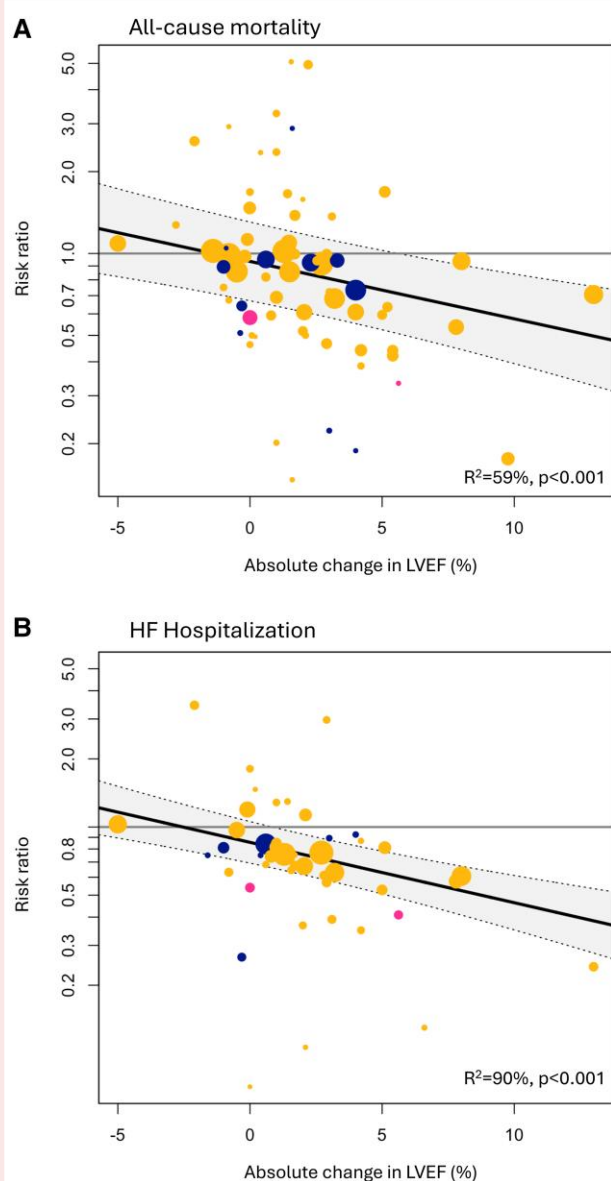


Figure 2 Correlations between treatment related effects on absolute change in left ventricular ejection fraction and treatment effects on (A) all-cause survival and (B) HF hospitalization. Yellow circles = HFrEF, blue circles = HFmrEF/HFpEF, and pink circles = mixed HFrEF and HFmrEF and HFmrEF/HFpEF. Each circle represents a trial, and the size of the circles is proportional to the inverse variance. Black straight lines represent weighted regression lines and dashed lines represent 95% prediction intervals. The grey straight line represents the line of unity where RR = 1.0

Patient reported outcomes and exercise capacity

Treatment related changes in MLWHFQ scores were strongly correlated with both risk of mortality (24 comparisons, $P < .001$, and $R^2 = 93\%$) and HF hospitalization (13 comparisons, $P < .0001$, and $R^2 = 100\%$) (Figure 4 and Table 1). The STE for MLWHFQ scores

was 2.6 point decrease for all-cause mortality (Table 1). Treatment related changes in KCCQ-OSS were weakly correlated with risk of all-cause mortality (14 comparisons, $P = .010$, and $R^2 = 36\%$) and not significantly associated with HF hospitalization (14 comparisons, $P = .184$, and $R^2 = 2\%$) (Figure 4 and Table 1; Supplementary Figure S5).

Findings were broadly consistent in multivariate analyses and when weighting by sample size (Supplementary Tables S4 and S5). However, there were too few trials with >1000 patient-years to assess for MLWHFQ scores in sensitivity analyses whilst findings for KCCQ-OSS were consistent with the primary analysis (Supplementary Table S6). Leave-one-out cross validation analyses were also broadly consistent (Supplementary Tables S7 and S8).

Change in 6MWT was moderately correlated with risk of all-cause mortality (33 comparisons, $P = .013$, $R^2 = 57\%$, and STE = 41 m) and weakly correlated with HF hospitalization (21 comparisons, $P = .027$, $R^2 = 38\%$, and STE = 44 m) (Figure 4 and Table 1). Among trials of HFrEF, change in 6MWT was strongly correlated with both all-cause mortality (21 trials, $P = .008$, $R^2 = 77\%$, and STE = 34 m) and HF hospitalization (13 trials, $P = .007$, $R^2 = 75\%$, and STE = +38 m), however this evidence was derived from small randomized clinical trials (Table 1; Supplementary Table S6). Change in peak VO_2 was not significantly associated with treatment related changes in mortality nor HF hospitalization (Figure 4 and Table 1).

Overall surrogacy assessment according to HF type

No biomarkers or echocardiographic measures were strong surrogates of all-cause mortality in HFrEF and HFmrEF/HFpEF (Graphic Abstract; Table 2). Treatment related change in LVEF was a strong surrogate for HF hospitalization in HFrEF. Change in LV end-systolic volume and LV end-diastolic diameter in HFrEF, and change in mitral annular e' in HFmrEF/HFpEF were graded as moderate surrogates for treatment effects on HF hospitalization. Change in 6 minute walk distance was also graded as a moderate surrogate for both mortality and HF hospitalization in trials of HFrEF with the main limitation being evidence that was derived from small randomized clinical trials that was not robust in sensitivity analyses.

Discussion

There is growing interest in incorporating surrogate markers into HF clinical trials to enhance statistical power and efficiency, reduce costs, and accelerate patient access to effective therapies. This meta-analysis of 96 randomized clinical trials, encompassing 120 304 participants, assessed the surrogacy of 15 candidate variables for all-cause mortality and HF hospitalization. Overall, evidence supporting the use of a single surrogate marker for mortality in HF is lacking (Graphic Abstract). Notably, NPs were only weakly correlated with treatment related effects on mortality and HF hospitalization. Changes in LVEF demonstrated moderate correlation with mortality and a strong correlation with HF hospitalization in trials of HFrEF. Patient reported outcomes, namely MLWHFQ scores were highly correlated with both mortality and HF hospitalization in the overall population but not separately by HFrEF and HFmrEF/HFpEF, reflecting that the evidence was derived from smaller clinical trials.

A prior meta-analysis of 16 trials found treatment related changes in NPs were not correlated with all-cause mortality but were modestly correlated with HF hospitalization.⁸ This study may have been underpowered to assess the true magnitude of association, and it did not calculate the minimum NP reduction required to impact clinical outcomes, nor assessed for differences in monitoring interval. Our study includes an additional 48 trials and a broader range of treatment effects, which increases analytic power. We found changes in

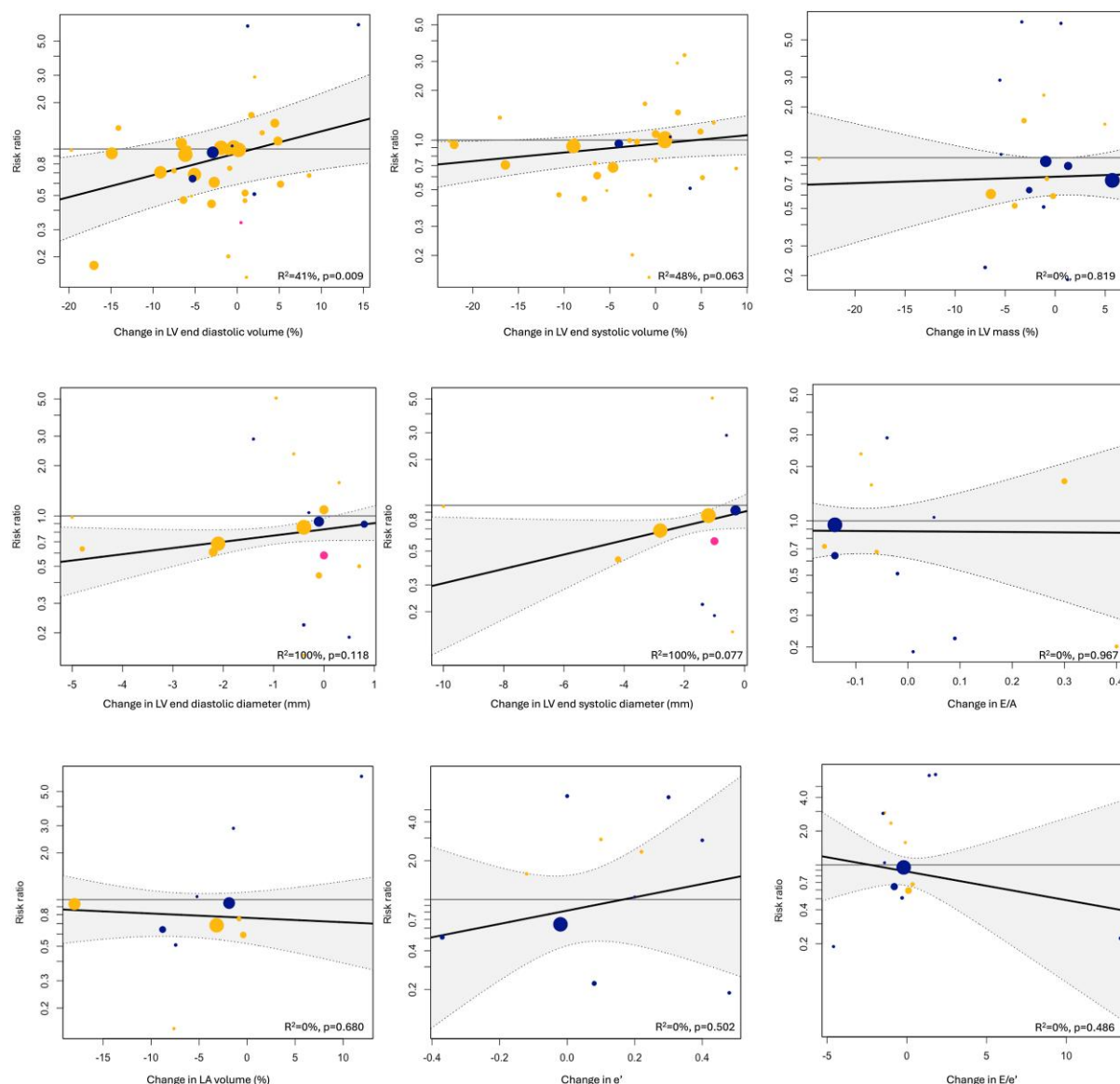


Figure 3 Correlations between treatment related effects on echocardiographic measures and all-cause survival. Yellow circles = HFrEF, blue circles = HFmrEF/HFpEF, and pink circles = mixed HFrEF and HFmrEF/HFpEF. Each circle represents a trial, and the size of the circles is proportional to the inverse variance. Black straight lines represent weighted regression lines and dashed lines represent 95% prediction intervals. The grey straight line represents the line of unity where $RR = 1.0$

NPs were weakly correlated with mortality and HF hospitalization, suggesting whilst NP reductions capture a portion of the treatment effect, they do not fully explain clinical outcomes. This reflects the complex physiology of HF progression and the influence of multiple factors on NP levels. Furthermore, NT-proBNP measurement exhibits up to 40% intra-individual variability without disease progression, complicating its utility as a surrogate.^{17–19} Changes in NPs at 6 months had the strongest association with clinical outcomes, consistent with a prior observational study that identified the minimal informative interval for NT-proBNP monitoring to be 7.9 months.²⁰ Notably, the present study was not intended to evaluate the prognostic significance of individual risk predictors, where achieved values of NT-proBNP may be more important than the change in risk marker.²¹

We found that changes in LVEF were moderately correlated with mortality and strongly correlated with HF hospitalization and the optimal assessment interval for change in LVEF was 12 months, with a 6% improvement identified as the surrogate threshold for mortality benefit. Practically, this means treatment related improvements in mean LVEF of at least 6% are needed in a phase 2 clinical trial to predict a significant reduction in mortality in larger phase 3 clinical trials. Importantly, LVEF should only be considered a surrogate in HFrEF trials as demonstrated in our subgroup analyses. In trials of HFrEF, echocardiographic markers of LV size, namely LV volumes and diameter were correlated with HF hospitalization but not mortality, but these data were less robust. In six trials of HFmrEF/HFpEF, treatment related changes in mitral annular e' were correlated with HF hospitalization but not mortality.

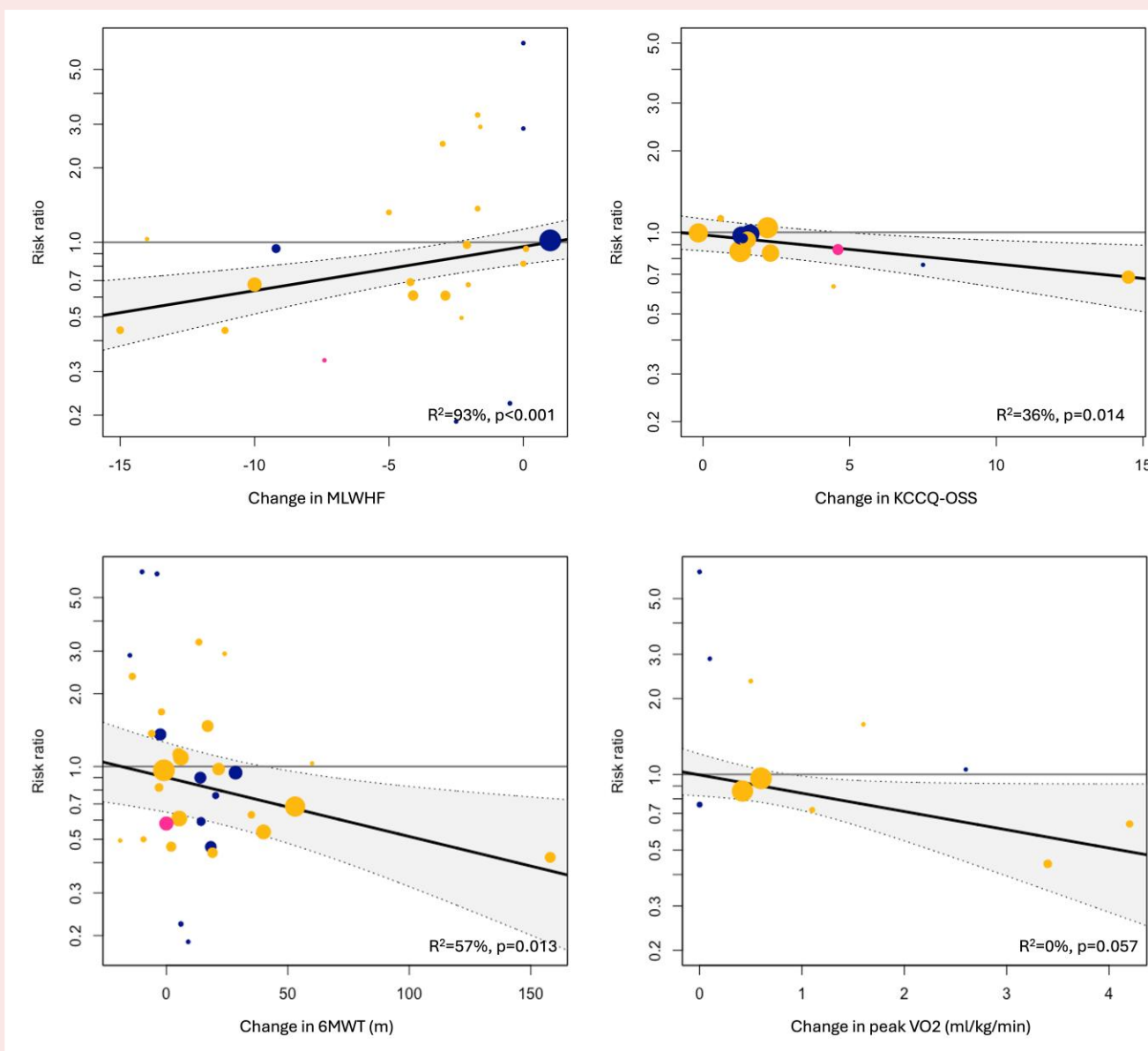


Figure 4 Correlations between treatment related effects on patient reported outcomes and exercise capacity with all-cause survival. Yellow circles = HFrEF, blue circles = HFmrEF/HFpEF, and pink circles = mixed HFrEF and HFmrEF/HFpEF. Each circle represents a trial, and the size of the circles is proportional to the inverse variance. Black straight lines represent weighted regression lines and dashed lines represent 95% prediction intervals. The grey straight line represents the line of unity where $RR = 1.0$

A prior study suggested exercise induced changes in 6MWT was a poor surrogate for mortality and HF hospitalization.⁹ However, this analysis was based on a limited number of trials and interventions that did not significantly reduce mortality, limiting its statistical power. We found treatment related changes in 6MWT were moderately correlated with risk reductions in mortality and weakly correlated with HF hospitalization. Although HF hospitalization may be an intermediary endpoint, it is possible that some surrogates are strongly associated with mortality but not HF hospitalization. This aligns with observational studies suggesting 6MWT is a strong predictor of mortality in HF and improvements in 6MWT are associated with increased survival.^{22–24} In our study, minimum improvement in 6MWT needed to observe a significant reduction in clinical endpoints was 40–45 m, a result consistent with some studies demonstrating that each 50 m increase in 6MWT is associated with improved survival.^{24,25}

Demonstration of benefit in hard endpoints such as mortality and HF hospitalization remain the gold standard for drug and device approval in HF. The US FDA have provided guidance that treatments could also be approved for improvements in symptoms, quality of life and functional capacity, but not based on biomarkers alone.²⁶ Our analyses support these conclusions, as treatment related change in NPs were weakly correlated with treatment related effects on mortality and HF hospitalization. Meanwhile, treatment related reductions in MLWHFQ scores were strongly correlated with reduced mortality and HF hospitalization in the overall population, although this was predominantly derived from analyses of smaller trials. Larger clinical trials are needed to more definitively assess the relationship between changes in MLWHFQ scores and hard clinical endpoints. Improvements in KCCQ-OSS had weak correlation with mortality and HF hospitalization, consistent with findings from prior HF trials that observed discrepancies in treatment effects on KCCQ and on the risk

Table 2 Surrogacy assessment of biomarkers, echocardiographic measures, patient reported outcomes, and exercise capacity for mortality and HF hospitalization stratified by trials of HFrEF and HFmrEF/HFpEF

	Surrogacy assessment	
	All-cause mortality	HF hospitalization
Biomarker		
Natriuretic peptides		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
Echocardiographic measures		
LVEF		
HFrEF	Limited/no evidence	Strong evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
LV end-diastolic volume		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
LV end-systolic volume		
HFrEF	Limited/no evidence	Moderate evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
LV end-diastolic diameter		
HFrEF	Limited/no evidence	Moderate evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
LV end-systolic diameter		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
LV mass		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
E/A ratio		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
LA volume		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
Mitral annular e'		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Moderate evidence
E/e'		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
Patient reported outcomes		
MLWHFQ score		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
KCCQ-OSS		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence

Continued

Table 2 Continued

	Surrogacy assessment	
	All-cause mortality	HF hospitalization
Exercise capacity		
6-minute walk test		
HFrEF	Moderate evidence	Moderate evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence
Peak VO ₂		
HFrEF	Limited/no evidence	Limited/no evidence
HFmrEF/HFpEF	Limited/no evidence	Limited/no evidence

Green = strong evidence, orange = moderate evidence, grey = limited/no evidence. KCCQ-OSS, Kansas City Cardiomyopathy Questionnaire Overall Summary Score; LA, left atrial; LV, left ventricular; LVEF, left ventricular ejection fraction; m, metres; MLWHFQ, Minnesota Living with Heart Failure Questionnaire; VO₂ oxygen uptake.

of mortality and HF hospitalization.²⁷ Notably, we identified the STE for KCCQ-OSS, wherein a ≥ 5 -point improvement was associated with a meaningful reduction in mortality risk. This reflects the change in KCCQ-OSS needed to be observed in a clinical trial to predict a significant reduction in all-cause mortality but does not necessarily equate to the minimal clinically important difference for KCCQ, which can range from 5 up to 20 points.^{13,27} The STEs identified in this study reflect the average changes in surrogate marker needed to predict a positive treatment effect at a trial level, and do not necessarily inform whether individual patients are responders or non-responders to treatment.

These analyses are based on trial-level estimates of change in surrogate markers and clinical endpoints. Therefore, the calculated correlations are likely to be a conservative estimate and attenuated in the presence of uncertainty surrounding the exact magnitudes of both sets of treatment effects. The population also applies to patients included in randomized trials, which limits the generalizability of the findings. Nevertheless, our primary objective was to assess surrogacy at the trial level to determine whether each marker captures a significant proportion of the treatment effect.^{3,4} Other surrogate markers (such as high sensitivity C-reactive protein, cardiac magnetic resonance imaging) of potential interest and associations with other meaningful clinical outcomes (such as all-cause hospitalizations) were not assessed because we were limited to data elements broadly reported in the included trials. Certain surrogate-clinical outcome pairs, such as peak VO₂ max and HF hospitalization, may have been underpowered. However, the lack of substantial trial-level evidence for these markers suggests they are not yet suitable as surrogates until further data emerge. Despite a comprehensive literature search, some studies may not have been captured. Finally, we were unable to account for the competing risk of death for HF hospitalization without access to the individual patient data.

In conclusion, there is insufficient evidence to support the use of a single surrogate marker for mortality in HF trials. NPs were weakly correlated with both mortality and HF hospitalization, whilst LVEF may be an appropriate surrogate for HF hospitalization in HFrEF. Although changes in MLWHFQ scores and 6 minute walk distance appear to have potential as surrogates for both all-cause mortality and HF hospitalization, this evidence is derived from smaller clinical trials. Future research into novel risk markers is needed to improve the efficiency of therapeutic developments in HF.

Author contributions

Nelson Wang (Data curation; Formal analysis; Methodology; Resources; Supervision; Writing—original draft; Writing—review &

editing), Thomas Hibbert (Data curation; Investigation; Methodology; Validation; Writing—review & editing), Sarah Ishak (Data curation; Investigation; Methodology; Writing—review & editing), Nikki Raftopoulos (Data curation; Investigation; Methodology; Writing—review & editing), Rita-Maria Abdo (Data curation; Investigation; Methodology), Christian Abhayaratna (Data curation; Methodology; Writing—review & editing), Phidias Rueter (Data curation; Methodology; Writing—review & editing), and Muthiah Vaduganathan (Conceptualization; Investigation; Methodology; Supervision; Writing—review & editing)

Supplementary data

Supplementary data are available at [European Journal of Heart Failure](#) online.

Declarations

Disclosure of Interest

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Data Availability

Data will be made available upon reasonable request.

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